

Contact Information

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🔗 ejpaboncancel.github.io

Education & Academic Background**Purdue University**

- Ph.D. in Mathematics | Advisor: Prof. Aaron Yip
- M.S. in Mathematics

West Lafayette, IN

Expected June 2029

May 2026

University of Puerto Rico, Mayagüez (UPRM)

- B.S. in Pure Mathematics, Minor in Applied Mathematics (*Magna Cum Laude*)

Mayagüez, PR

June 2023

Skills and Other InformationProgramming & Computation: Python, Julia, C++, SageMath, MATLAB | Formatting & Tools: HTML, $\LaTeX$, Git

Math & AI/ML: NumPy, SciPy, PyTorch, TensorFlow, JAX, Flux, CUDA | Spoken Languages: Fluent in English and Spanish

Research Experience**URA-Sandia Graduate Student Summer Fellow**, URA-Sandia National Laboratories

May 2025–August 2025

- Studied machine learning surrogate models, and learned and applied data assimilation for closure models with JAX.
- Conducted a parametric study of the optimization step for 100 noisy samples of an SIQR epidemic model.
- Determined a minimum ensemble size of 500 ensembles for optimal Ensemble Kalman filter-based data assimilation.

GEM Fellow Summer Research Program Intern, MIT Lincoln Laboratory

May 2023–August 2023

- Developed mathematical algorithms for autoencoders with LSTM architecture, using the Flux library.
- Identified the autoencoder that minimized the loss function by using PCA and clustering techniques.
- Used K-medoids to obtain an optimal silhouette score of 0.875, and IF-1 score of 0.839.

Papers and Articles

- [1] S. Polk, E.J. Pabón Cancel, R. Paleja, K. Chestnut-Chang, R. Jensen, and M. Ramírez. Unsupervised Network-Based Behavior Inference from Human Action Sequences (UNBIAS). *2024 IEEE Conference on Games (CoG) 2024*, pp. 1-8.
- [2] D. Chen, P.E. Harris, J. Carlos Martinez Mori, E.J. Pabón Cancel, and G. Sargent. Permutation Invariant Parking Assortments. *Enumerative Combinatorics and Applications*, **4:1**, 1-25 (2024). #S2R4.
- [3] I. Byrne, N. Dodson, R. Lynch, E.J. Pabón Cancel, and F. Piñero-González. Improving the Minimum Distance Bound of Trace Goppa Codes. *Designs, Codes and Cryptography*. **91**, 2649–2663 (2023).

Projects**Project in Educational Topic (TBD)**, URA STEM Ambassadors Program

May 2026–December 2026

2026 URA STEM Ambassadors Challenge

TBD

Project in Biotechnology (3rd Place Winner), MIT Lincoln Laboratory Summer Research Program

June 2023–July 2023

2023 MIT Lincoln Laboratory Intern Innovative Idea Challenge (I³C) | Mentors: Ryan Burrow and Ashok Kumar*SKINS: Skin-growth boosting and Intra-absorptive Solution bandages***Awards and Merits**

2025 Universities Research Association-Sandia Graduate Student Summer Fellowship

May 2025–August 2025

2023 National GEM Consortium PhD Science Fellowship

August 2023–May 2024

- Purdue University Department of Mathematics Sponsorship

August 2023–May 2024

- MIT Lincoln Laboratory Employer Sponsorship

May 2023–August 2023

2023 MIT Lincoln Laboratory I³C 3rd Place Research Proposal Prize

July 2023

2023 Ford Foundation Predoctoral Fellowship Honorable Mention

March 2023

2022 Evertec Inc. STEM Scholarship

October 2022

2022 Hispanic Scholarship Fund Scholar

June 2022

National Math Alliance Predoctoral Scholar

November 2021

Puerto Rico-Louis Stokes Alliance for Minority Participation Research Scholarship

August 2019–December 2022

UPRM Faculty of Arts and Sciences Honor Roll

August 2018–May 2023

Eagle Scout Rank, with two Silver Palms

May 2017